

TrES-3b

R-1378

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_200325 · created 2026-08-01T01:38:07+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2458933.8958 +/- 0.0015 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.109 +/- 0.037
Transit depth (Rp/Rs)^2	1.18 +/- 0.8 [%]
Orbital Inclination (inc)	82.39 +/- 1.43
Airmass coefficient 1 (a1)	0.9704 +/- 0.0096
Airmass coefficient 2 (a2)	0.0199 +/- 0.0048
Scatter in the residuals of the lightcurve fit is	0.99 %
Best Comparison Star	#3 - [617, 338]
Optimal Method	PSF photometry
Transit Duration (day)	0.05 +/- 0.016

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R ² fit vs published	0.109 ± 0.037 vs 0.16309 (deviation 1.27σ)
Duration fit vs published	0.05 d vs 0.0567 d (deviation 0.36σ)
Mid-time O-C	-4.4 min
Depth SNR	2.6
Residual scatter	0.99 %

Light curve

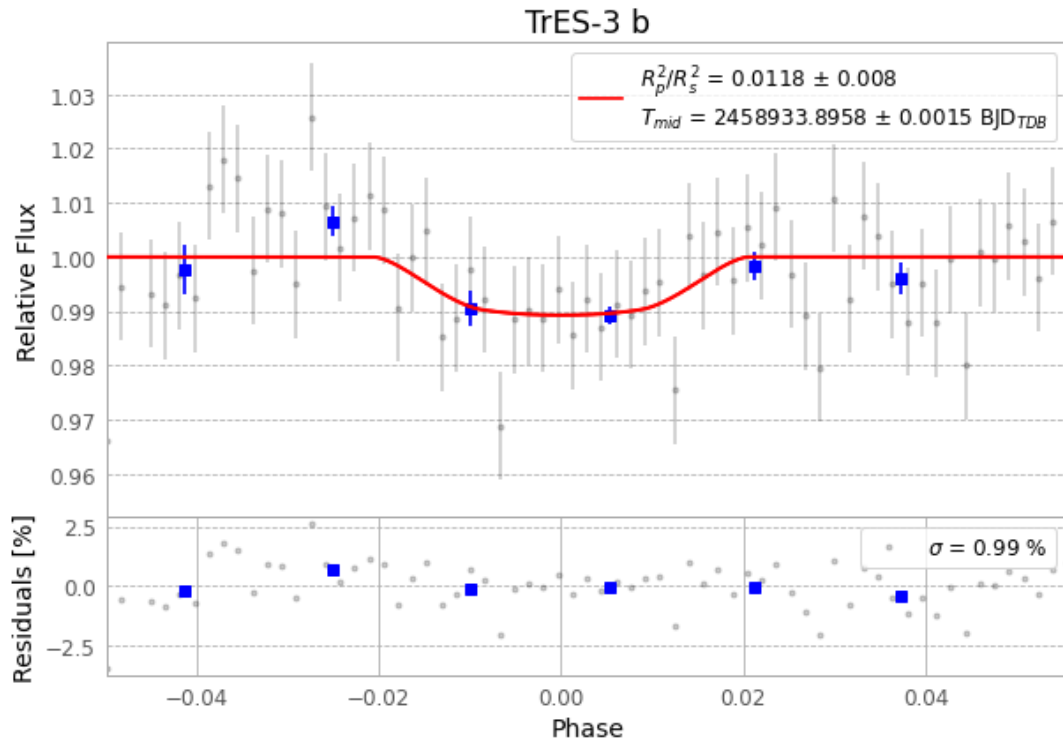


Figure 1 — FinalLightCurve_TrES-3 b_2020-03-25.png (SHA-256 1c07f2d422a14e18...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [511, 180], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 de9f99f43e89f6cb...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

67 FITS frames (44 MB), TRES-3200325075412.FITS → TRES-3200325111212.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-200325.log (ab517bfddb55...), FinalLightCurve_TrES-3 b_2020-03-25.png (1c07f2d422a1...), FinalLightCurve_TrES-3 b_2020-03-25.csv (867ec8c5de00...)

Compute resources

Wall time 125.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 01:38Z.